Learned Teacher-Supervised EHR Context Selection to Improve Small LLM Clinical QA

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Project Overview

Headline: *Efficient EHR QA via learned evidence selection.*
Claim: Competitive with semantic RAG while selecting **more structurally relevant** context, improving interpretability and efficiency.

- **Clinical QA needs patient-specific evidence**, but EHRs are long and messy—small models can't fit enough context and heuristic retrieval often misses what matters.
- We study **learned evidence selection under a strict** context budget to make answers more reliable and efficient.
- **Result:** A simple learned selector is competitive with semantic RAG and improves interpretability by exploiting document structure, returning top k chunks to feed into a smaller LLM with a limited context window

Datasets & Metrics

Datasets

- **EHR source (MIMIC-IV)**: De-identified hospital data with discharge summaries (text) + timestamped events (labs, vitals, meds, diagnoses, procedures).
- Training QA (teacher-generated): A larger LLM generates question–answer pairs over each patient timeline (scalable supervision).
- Holdout benchmark (gold): **EHRNoteQA v1.0.1** — 962 clinician-reviewed QA pairs over discharge summaries, used only for evaluation.

Metrics

- **Recall@K (retrieval):** With only K chunks of context, what fraction of answer-supporting evidence is retrieved? (K = 1, 3, 5, 10).
- **Token F1 (**answer overlap):
- **ROUGE-L** (sequence match): Rewards longest matching subsequences
- LLM-as-judge (1–5): A rubric score for correctness + completeness vs the gold answer; we report mean score and win rate across retrieval strategies.

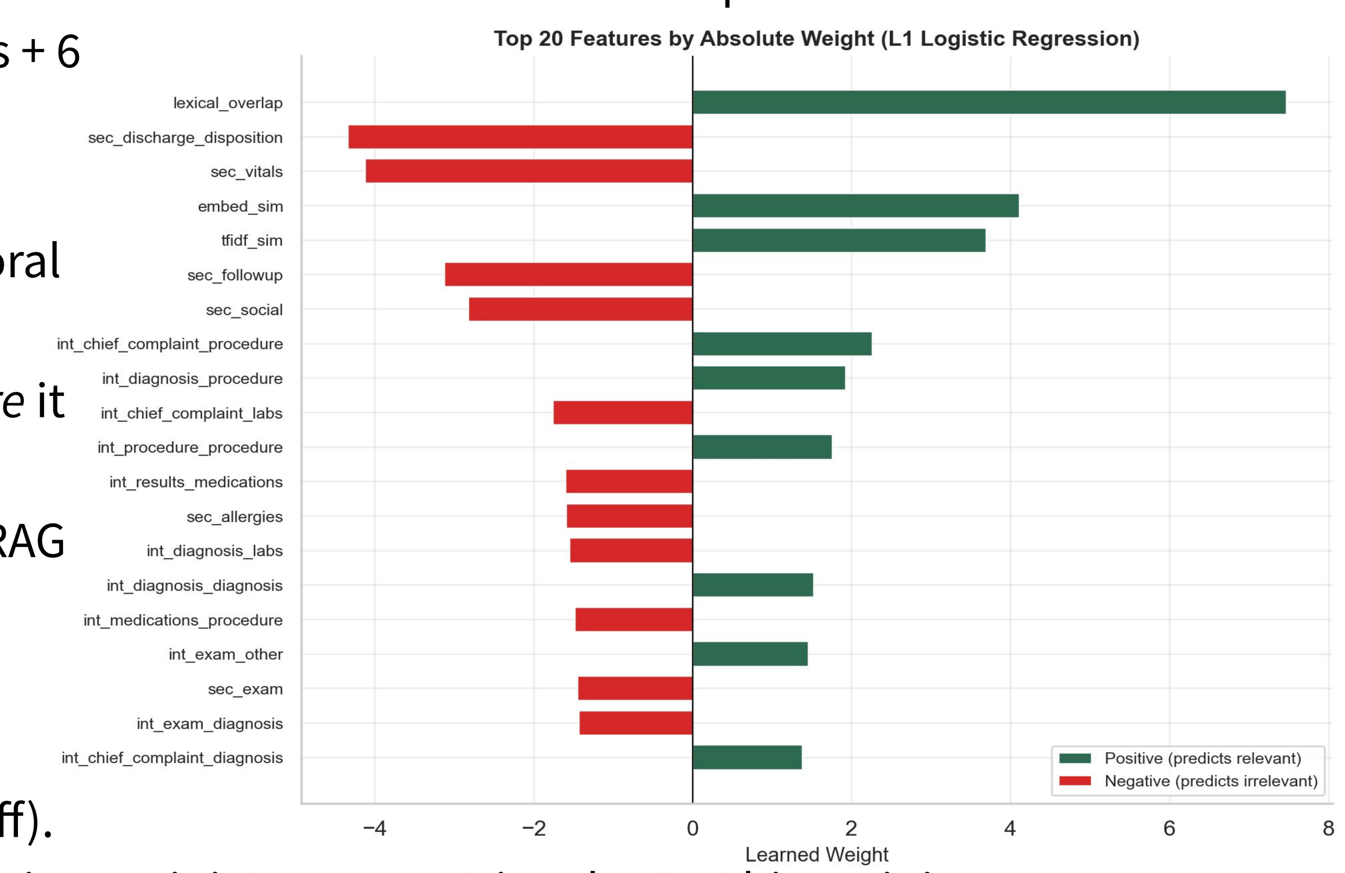
References

- Kweon et al. *EHRNoteQA: An LLM Benchmark for Real-World Clinical Practice Using Discharge Summaries* (PhysioNet v1.0.1, 2024). doi:10.13026/acga-ht95
- *MIMIC-IV-Note: Deidentified free-text clinical notes* (PhysioNet v2.2). https://physionet.org/content/mimic-iv-note/2.2/
- Kotschenreuther. *EHR-DS-QA: A Synthetic QA Dataset Derived from Medical Discharge Summaries* (PhysioNet v1.0.0, 2024). https://physionet.org/content/ehr-ds-qa/1.0.0/
- Wong et al. *Retrieval-augmented systems can be dangerous medical communicators* (arXiv:2502.14898, 2025). https://arxiv.org/abs/2502.14898

Methods & Experiments

Methods:

- Build patient timelines: Convert each admission into a sequence of candidate evidence chunks
- Learned selector: Train a sparse logistic regression ranker to score (question, chunk) pairs using:
  - similarity signals (TF-IDF + embeddings)
  - document structure (section + question-type interactions)
  - temporal / clinical salience cues
- Inference: For each question, select top-K chunks under a token budget and feed only those to a frozen small LLM ("student").
- L1 Logistic regression model: $\mathbf{p(useful = 1 \mid q, c) = \sigma(w^T \phi(q, c)),\ \ \sigma(z) = 1 / (1 + e^{-z})}$
- **Feature vector:** $\phi(q,c)$ is **166 dimensions** (per question–chunk pair)
- **Similarity (match):** embedding cosine + TF-IDF cosine + token overlap
- **Structure (where):** 19 section flags + 6 question-type flags + 102 section×question interactions
- **Clinical cues (importance):** temporal language, labs, sections
- **Why:** captures *what* matches, *where* it appears, and *clinical importance*—more than similarity RAG
- Report: three metrics
- Efficiency focus: Compare performance vs average context tokens (accuracy–efficiency tradeoff).



- Repeat with various models (o4-mini, 4o-mini, Lama 3.1, Mistral 7-B, Phi-3-mini

Discussions & Future Research

Discussions:

- **Not just semantic similarity:** The model learns document structure (section + section×question type) as a dominant signal for relevance, improving interpretability.
- **Why this helps:** Clinical notes are messy; key facts appear in lists/abbreviations (e.g., "mg", "bid"), so structure-aware selection can beat "similarity-only" retrieval in some cases.
- **Metrics tell different stories:** Token-overlap metrics can miss paraphrases and penalize list omissions; LLM-as-judge better captures correctness/completeness when context is more coherent.
- **Main limitation:** Training labels are weak (overlap-based evidence proxy), and teacher-generated QA can contain bias/omissions → need stronger gold supervision.
- **Efficiency story:** Comparable answer quality with fewer context tokens than full-context baselines (see efficiency–accuracy plot).

Future Research:

- **Better supervision:** more clinician-reviewed QA and ideally evidence-linked annotations (which chunk supports the answer).
- **Richer temporal modeling:** Trends and change-points in labs ; event-type-specific time windows
- **Stronger rankers:** Two-stage retrieval + reranking (MLP/cross-encoder) while keeping budgets fixed.
- **Coherence constraints:** Select contiguous note windows / coherent time windows to reduce scattered context.

Results

- **Competitive with semantic RAG:** Our learned selector performs similarly overall and wins in some settings, especially under tight budgets, while offering a structural alternative, in cases when RAG fails (redundancy)
- **Strong evidence retrieval:** Recall@K rises quickly with K (baseline selector: @1 0.45 → @3 0.71 → @5 0.84 → @10 0.96).
- **Structure beats similarity:** The selector relies heavily on section + section×question-type features (not just embedding similarity), improving interpretability.
- **Judge agrees with mechanism:** LLM-as-judge often shows clearer gains than token overlap metrics, consistent with more coherent, relevant context.